SystemCraft – AI Engineer – Production-Ready Generative AI

Course Overview

SystemCraft is a structured, production-oriented learning roadmap designed to teach how **real-world Generative AI software systems are engineered, built, and deployed**.

This course does not focus on mathematical theory or research-level model training.
Instead, it emphasizes **engineering clarity, system design, and practical implementation**.

The roadmap is released **incrementally in phases** to avoid learner overload and to ensure steady, confidence-driven progress.

Target Audience

This course is intended for:

- Beginners seeking a clear and structured entry into Gen-AI engineering
- Backend or full-stack developers transitioning into AI systems
- Engineers who want to build production-ready AI applications
- Learners who prefer system-level understanding over isolated tutorials

Phase 1 – Foundations

Objective

Establish clarity, engineering mindset, and core programming foundations required for Gen-AI systems.

Level 0 – Gen-AI Engineer Mindset

This level focuses on conceptual clarity before implementation.

Topics covered:

- Role and responsibilities of a Gen-AI engineer
- Differences Differences between AI demos and real production systems
- How Generative AI integrates into backend architectures
- Common failure points in real-world AI products
- Course structure, learning path, and progression

Level 1 – Python for Gen-AI Systems (Focused)

This level covers **only the Python concepts required for AI backend development**.

Topics covered:

- Python fundamentals for backend and AI workflows
- Writing clean and maintainable functions
- Working with JSON and API data
- Practical object-oriented programming:
- Asynchronous programming fundamentals (`async` / `await`)

Level 2 – Backend Fundamentals with FastAPI

This level introduces backend development principles required for AI systems.

Topics covered:

- API fundamentals and request–response lifecycle
- FastAPI project structure and architecture
- Data validation using Pydantic models
- Error handling and response management
- Integrating backend services with AI logic

Phase 2 – Backend Systems

Objective

Transform backend services into persistent, production-capable systems.

Level 3 – Backend Systems with Database (FastAPI + PostgreSQL)

Topics covered:

- Why databases are essential for Gen-AI applications
- PostgreSQL fundamentals for application developers
- Schema design and relationships
- CRUD operations
- SQLAlchemy ORM basics
- Clean architecture patterns (service and repository layers)

Phase 3 – Core Generative AI Engineering

Objective

Develop a deep, practical understanding of how Gen-AI systems work and how to control them.

Level 4 – Core Generative AI Concepts (Engineering Perspective)

Topics covered:

- What Large Language Models are from a systems perspective
- Transformer architecture explained conceptually
- Tokens, context windows, and cost implications
- Causes of hallucinations and reliability issues
- Prompt design patterns and best practices

Level 5 – Building Generative AI Applications

Topics covered:

- Safely integrating LLM APIs
- Designing conversational AI systems
- Context and memory management
- Retrieval-Augmented Generation (RAG)
- Embeddings and vector search fundamentals
- Logging, retries, and rate limiting

Phase 4 – Frontend Integration

Objective

Enable deployment of usable AI products with minimal frontend complexity.

Level 6 – Frontend for Generative AI Products

Topics covered:

- JavaScript fundamentals required for API consumption
- React basics for AI-driven interfaces
- Next.js fundamentals
- Streaming AI responses to the UI
- Secure frontend-backend communication

Phase 5 – Production and Cloud

Objective

Prepare systems for real-world usage, scale, and operational stability.

Level 7 – Production-Ready Gen-AI Systems

Topics covered:

- Scaling considerations for AI systems
- Latency and performance optimization
- Security fundamentals for AI applications
- Environment configuration and secrets management
- Cost control strategies

Phase 6 – Capstone Gen-AI Projects

Objective

Apply all previously learned concepts to design, build, and deploy **real-world, production-grade Generative AI systems**.

This level focuses on **end-to-end engineering**, not isolated features.

Projects Built

Project 1 – Production-Ready AI Chat Application

- Multi-user AI chat system
- Backend built with FastAPI
- Conversation persistence using PostgreSQL
- Context and memory management
- Cost-aware LLM usage
- Basic authentication and user isolation

Project 2 – Document-Based AI Assistant (RAG System)

- Upload and process documents (PDFs, text)
- Chunking and embedding pipelines
- Vector search and retrieval
- Retrieval-Augmented Generation (RAG)
- Source-grounded AI responses
- Metadata storage and query tracking

Project 3 – AI-Powered SaaS Application

- Full-stack Gen-AI product
- User authentication and authorization
- AI feature exposed as a SaaS capability
- Backend usage tracking and limits
- Clean frontend integration (Next.js)
- Environment-based configuration

Project 4 – Real-Time AI System (Streaming Responses)

- Streaming LLM responses to frontend
- WebSocket or server-sent events integration
- Low-latency response handling
- Error recovery and partial responses
- Real-time UX considerations

Engineering Focus Areas

Across all projects, you will apply:

- System architecture and clean layering
- Production-grade error handling

- Logging and monitoring
- Cost control strategies
- Security and data isolation
- Performance and scalability considerations

Learning Outcomes

By the end of this level, you will be able to:

- Design end-to-end Gen-AI systems
- Build AI applications beyond demos
- Debug and optimize production AI workflows
- Explain system architecture confidently
- Continue learning independently with a strong foundation